Specialization

John works at a hospital. He is an employee with attributes EMP_ID, EMP_NAME, EMP_DEPT, and EMP_SALARY. He works both as a doctor and as a researcher. As a doctor, he has attributes like MED_SPECIALTY, and as a researcher, he has attributes like RES_FIELD. He has a manager and a mentor who are his supervisors.

At a car rental company, a vehicle has attributes VIN, MODEL, YEAR, and COLOR. A vehicle can be either a car or a truck, but not both. Cars have attributes like DOOR_COUNT and FUEL_TYPE, and trucks have CARGO_CAPACITY and AXLE_COUNT. The company also rents out motorcycles as a separate subtype.

At a hospital, an employee has EMP_ID, FULL_NAME, ADDRESS, and HIRE_DATE. He/she may work as a doctor, a nurse, or a technician. An employee can hold any combination of these three roles. Doctors have SPECIALTY and LICENSE_NO, nurses have SHIFT and WARD, technicians have CERTIFICATION and EQUIP_TYPE. Employee has a spouse (if married) and any number of children as dependents, but the spouse and children are not employees.

A vehicle database tracks VEH_ID, LICENSE_PLATE, MAKE, MODEL, and YEAR. Vehicles can be either owned by a person or owned by a company. Privately owned vehicles have OWNER_SSN and PARKING_TYPE, company-owned vehicles have COMPANY_ID and FLEET_NUMBER. All vehicles are classified as either electric or normal (petrol or diesel), and an electric vehicle has BATTERY_CAPACITY, while a normal vehicle has FUEL_TANK_SIZE. Electric vehicles can be privately or company-owned.

A bank account has ACC_NO, BALANCE, OPEN_DATE, and STATUS. Accounts are either personal or business. Personal accounts are further divided into checking and savings accounts, but a personal account can be both a checking and a savings. Business accounts are either merchant or corporate. Checking accounts have OVERDRAFT_LIMIT, savings have INTEREST_RATE. Merchant accounts have TRANSACTION_FEE, and corporate accounts have CONTACT_PERSON.

On a construction site, an asset has ASSET_ID, DESCRIPTION, PURCHASE_DATE, and COST. Assets are either equipment or material. Equipment can be heavy machinery or hand tools. Heavy machinery has OPERATOR_CERT_REQ and FUEL_TYPE, hand tools have MATERIAL_TYPE. Materials are either raw or prefabricated. Workers have WORKER_ID, NAME, and PHONE. A worker can operate multiple pieces of heavy machinery and be assigned to many materials.

A product has PROD_ID, NAME, PRICE, and STOCK. Products are physical goods, digital goods, or services. Physical goods have WEIGHT and DIMENSIONS, digital goods have DOWNLOAD_LINK and LICENSE_KEY, services have DURATION and PROVIDER. Physical goods are further classified as electronics, clothing, or furniture. An item of clothing can also be a returned item with RETURN_REASON and REFUND_STATUS.

A staff member has EMP_ID, NAME, PROJECT, and TITLE. Staff can be developers and testers. Developers have PROGRAMMING_LANG and CERTIFICATION, testers have TEST_TOOL and BUG_TRACKER. Any staff member can be a team lead. Team leads have LEAD_START_DATE and TEAM_SIZE. Some developers are also testers.

A shipment has SHIP_ID, WEIGHT, DESTINATION, and STATUS. Shipments are either priority or standard. Priority shipments have RUSH_FEE and DELIVERY_WINDOW, standard have ESTIMATED_DAYS. Carriers have CARRIER_ID, NAME, and CAPACITY. Carriers can be truck, air, or rail. A shipment is assigned to exactly one carrier, but a carrier can handle multiple shipments. Some carriers are also subcontractors. Subcontractors have CONTRACT_ID and RATE.